

# Web Technologies Project Report: Tary Ethno Cafe (Astana Flagship)

Course: Web Technologies / Frontend Fundamentals

Repository URL: <https://github.com/jasulansam/web>

## Team Members:

- Zamrat Zhasulan
- Zhaksylyk Adilet

## 1. Project Overview

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- **Objective:** Development of a multi-page website for the Tary Ethno Cafe flagship branch (Kabanbay Batyr 15/1, Astana) adhering strictly to semantic HTML5 standards without external styling or client-side scripting frameworks.
- **Core Architectural Rules:**
  - Strict landmark hierarchy (<header>, <nav>, <main>, <footer>).
  - Pure semantic structure (no inline styling, zero presentation attributes like border="1" or align).
  - Tabular data formatting using accessible table structures (<table>, <caption>, <thead>, <tbody>, and th scope).
  - Form inputs configured with explicit label associations, logical field grouping (<fieldset>, <legend>), and native HTML5 validation constraints.

## 2. Task A — Anatomy of a Real Page (Reference Audit: coffeeboom.kz)

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### 2.1 Head Elements and Metadata Inspection

- **DOCTYPE:** <!DOCTYPE html>
- **Language:** lang="kk" in <html class="html" lang="kk">
- **Charset:** Legacy HTTP-EQUIV syntax <meta http-equiv="Content-Type" content="text/html; charset=utf-8"/>
- **Viewport:** <meta name="viewport" content="width=device-width, initial-scale=1.0">
- **Title:** Coffee Boom – Қазақстан, Ресей және Өзбекстандағы кофеханалар желісі.
- **Three Key Meta Tags:**
  1. <meta name="Description" content="Coffee Boom – 2010 жылдан бері жұмыс істеп келе жатқан, 130-дан астам кофеханасы бар халықаралық желі...">
  2. <meta name="theme-color" content="whitesmoke">
  3. <meta property="og:image" content="https://coffeeboom.kz/wa-data/public/wcms/images/07/04/00/461/461.0x0.png">

### 2.2 DevTools Source Inspection (Div-Soup Demonstration)

Inspect Element - coffeeboom.kz (Div-Soup & Legacy Metadata)

```
<!DOCTYPE html>
<html class="html" lang="kk">
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=utf-8"/>
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Coffee Boom – Қазақстан, Ресей және Өзбекстандағы кофеханалар желісі.</title>
    <meta name="Description" content="Coffee Boom – 2010 жылдан бері жұмыс істеп келе жатқан...">
  </head>
  <body>
    <div class="site-wrapper">
      <div class="header-box"> <!-- Div used instead of <header> -->
        <div class="nav-menu-bar"> <!-- Div used instead of <nav> -->
          <div class="menu-item"><a href="/about">Біз туралы</a></div>
          <div class="menu-item"><a href="/franchise">Франшиза</a></div>
          <div class="menu-item"><a href="/roastery">Кофе фабрикасы</a></div>
        </div>
      </div>
      <div class="main-content-area"> <!-- Div used instead of <main> -->
        <div class="slider-container">...</div>
        <div class="cards-grid"> <!-- 570 generic <div> tags total -->
```

| Element Type        | Tag Name  | Occurrence Count | Status / Evaluation                                   |
|---------------------|-----------|------------------|-------------------------------------------------------|
| Semantic Header     | <header>  | 1                | Present for brand banner                              |
| Semantic Navigation | <nav>     | 0                | Severe issue: navigation wrapped only in generic divs |
| Sectional Grouping  | <section> | 0                | No thematic section tags used                         |
| Independent Article | <article> | 0                | Blog/magazine articles use plain divs                 |
| Complementary Aside | <aside>   | 0                | No peripheral landmarks                               |
| Data Tables         | <table>   | 0                | No structured tables for menu or branch data          |
| Generic Containers  | <div>     | 570              | Excessive div-soup throughout entire layout           |

### 3. Three Structural Mistakes and Our Solutions

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#### 1. Extreme "Div-Soup" and Missing Navigation Landmarks:

- *Mistake:* On `coffeeboom.kz`, the main navigation bar containing over 20 links is wrapped in generic `<div>` tags rather than a semantic `<nav>`. Screen readers and search engine crawlers cannot identify navigation landmarks.
- *Our Solution:* On all our pages (`index.html`, `menu.html`, `booking.html`, `colophon.html`), site navigation is strictly encapsulated in a semantic `<nav>` tag holding a structured `<ul>` list.

#### 2. Outdated Legacy Charset Declaration:

- *Mistake:* The site uses the obsolete HTML4 syntax `<meta http-equiv="Content-Type" content="text/html; charset=utf-8"/>`.
- *Our Solution:* We implemented the modern, concise HTML5 standard `<meta charset="UTF-8">` on every document.

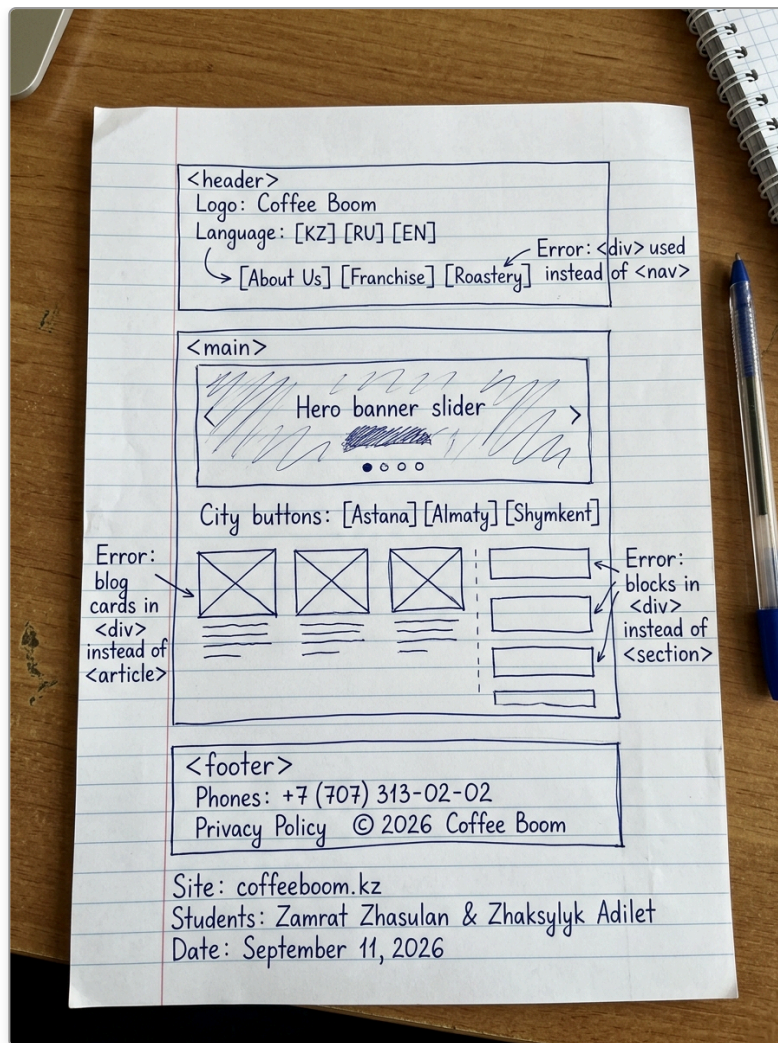
#### 3. Absence of Accessible Tables for Tabular Information:

- *Mistake:* The real website displays regional branches, locations, and menu pricing via unsemantic floating divs (zero `<table>` elements), making data extraction inaccessible for assistive tools.
- *Our Solution:* In our files (specifically `menu.html` for pricing), we implemented an accessible `<table>` complete with `<caption>`, `<thead>`, `<tbody>`, and explicit scopes (`scope="col"`, `scope="row"`).

### 4. Hand-Drawn Page Structure Sketch

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As required by Assignment 1, the page structure of the audited reference website was analyzed and hand-drawn on paper with pen by the students, highlighting semantic errors, landmark structure, and signed by both authors.



## 5. Task B — Reflective Essays

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### Reflective Essay: Zamrat Zhasulan

A web page is fundamentally a plain-text document structured with HyperText Markup Language (HTML). At its core, an HTML file consists of human-readable elements demarcated by tags, attributes, text nodes, and entities. When a user navigates to a local HTML file or web address, the browser receives a continuous stream of raw bytes.

The rendering engine decodes these bytes into characters using the declared character encoding (UTF-8), tokenizes the stream into distinct start tags, end tags, and content, and constructs the Document Object Model (DOM) tree. Concurrently, the engine parses external stylesheets to generate the CSSOM (CSS Object Model). The browser merges DOM and CSSOM into a Render Tree, filtering out non-rendered nodes (such as `<head>` or elements with `display: none`). Next, the layout engine executes geometry calculations to determine the precise size and coordinates of each element on the viewport. Finally, the painting stage rasterizes vectors, text, borders, and colors onto physical pixels displayed on the screen. Because our assignment strictly excludes CSS and JavaScript, the browser renders directly using its default User-Agent stylesheet.

In my codebase, I strictly prioritized semantic markup over generic `<div>` containers:

1. In `menu.html`, I wrapped our pricing data inside an accessible `<table>` with `<thead>` and `<tbody>` instead of styled grid divs, giving assistive screen readers clear tabular data relationships.
2. In `menu.html`, I utilized a definition list (`<dl>`, `<dt>`, `<dd>`) instead of divs to define traditional ingredients (Tary, Balkaymak, Kurt) as term-and-description semantic pairs.
3. In `index.html`, I chose `<article>` instead of `<div>` for the cafe heritage section because it forms an independent, self-contained story that can stand on its own.

### Reflective Essay: Zhaksylyk Adilet

A web page is a structured text document containing semantic tags that define document landmarks, interactive forms, and content hierarchies. When a browser opens our `booking.html` file, it parses the markup from top to bottom, building an in-memory tree representation known as the DOM (Document Object Model). Without external CSS, the browser applies its built-in user-agent style sheet to compute font sizes, element margins, and form control layouts before painting pixels to the display.

In my codebase, I strictly chose semantic elements over non-semantic `<div>` tags:

1. In `booking.html`, I grouped personal contact details and dining preferences inside `<fieldset>` elements with `<legend>` rather than wrapper divs, ensuring screen readers announce logical category contexts.
2. In `booking.html`, I explicitly connected each `<label for="...">` with its corresponding input id instead of using generic divs with text, ensuring clicking labels activates input focus.
3. In `colophon.html`, I used `<pre>` and `<code>` instead of a div container to display sample HTML code while preserving whitespace and formatting.

When a visitor clicks the submit button on `booking.html` today, the browser triggers native client-side constraint validation against attributes like `required`, `type="email"`, `type="number"`, and `type="tel"`. If any input fails validation, the browser blocks submission and displays a native alert. Because no server-side backend script processes the request, the browser attempts an HTTP POST request to `action="#"`, reloading the static page with query fragment without transmitting data over the network.

## 6. AI Assistance & Usage Logs

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Interactive logs documenting prompt history, academic integrity declarations, and AI-assisted workflow iterations are maintained in the GitHub repository:

- Zamrat Zhasulan — AI Usage Log: <https://github.com/jasulansam/web/blob/main/ai-log.md>

- Zhaksylyk Adilet — AI Usage Log: <https://github.com/jasulansam/web/blob/main/ai-log.md>

## 7. W3C Validation Evidence

All four project documents were submitted to the official W3C Nu HTML Validator, passing with zero errors and zero warnings.

### 1. val-index (index.html — Passed, 0 Errors)

### Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for contents of text-input area (checked with vnu 26.9.9)

Show: ☒ source ☐ outline ☐ image report | Options: ☐ errors & warnings only | Check by: text input

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Official website of Tary Ethno Cafe in Astana, Kazakhstan, celebrating authentic Kazakh gastronomic tr
  <meta name="author" content="Jasulan">
  <!-- Author: Jasulan -->
  <title>Tary Ethno Cafe Astana - Authentic Kazakh Coffee & Heritage</title>
</head>
```

Document checking completed. No errors or warnings to show.

Used the HTML parser.  
Total execution time 4 milliseconds.

[About this checker](#) • [Report an issue](#)

### 2. val-menu (menu.html — Passed, 0 Errors)

### Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for contents of text-input area (checked with vnu 26.9.9)

Show: ☒ source ☐ outline ☐ image report | Options: ☐ errors & warnings only | Check by: text input

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Discover the artisanal menu at Tary Ethno Cafe Astana, offering signature millet beverages, steppe tea
  <meta name="author" content="Jasulan">
  <!-- Author: Jasulan -->
  <title>Menu & Traditional Offerings - Tary Ethno Cafe Astana</title>
</head>
```

Document checking completed. No errors or warnings to show.

Used the HTML parser.  
Total execution time 4 milliseconds.

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### 3. val-booking (booking.html — Passed, 0 Errors)

## Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for contents of text-input area (checked with vnu 26.9.9)

Show: ☒ source ☐ outline ☐ image report | Options: ☐ errors & warnings only | Check by: text input

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Reserve a table at Tary Ethno Cafe in Astana for authentic tea ceremonies, specialty coffee tastings, and live music performances.">
  <meta name="author" content="Jasulan">
  <!-- Author: Jasulan -->
  <title>Table Reservations & Inquiries - Tary Ethno Cafe Astana</title>
</head>
```

Document checking completed. No errors or warnings to show.

Used the HTML parser.  
Total execution time 4 milliseconds.

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### 4. val-colophon (colophon.html — Passed, 0 Errors)

## Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for contents of text-input area (checked with vnu 26.9.9)

Show: ☒ source ☐ outline ☐ image report | Options: ☐ errors & warnings only | Check by: text input

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Colophon and technical documentation explaining how the Tary Ethno Cafe website was constructed using modern web technologies.">
  <meta name="author" content="Jasulan">
  <!-- Author: Jasulan -->
  <title>Colophon & Technical Architecture - Tary Ethno Cafe</title>
</head>
```

Document checking completed. No errors or warnings to show.

Used the HTML parser.  
Total execution time 4 milliseconds.

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## 8. HTML5 Semantic Elements Checklist

The comprehensive line-by-line checklist of every required HTML5 semantic tag, attribute, list, and table structure with student attributions is documented in the repository:

URL: <https://github.com/jasulansam/web/blob/main/tag-checklist.md>